MIL-C-5/9F
12 AUGUST 1966
SUPERSEDING
MIL-C-5/9E
21 December 1960

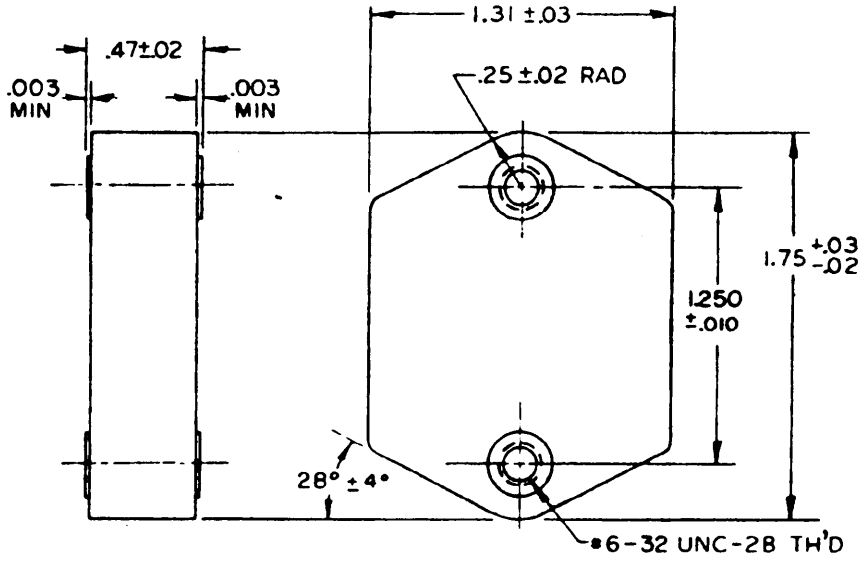
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC,

STYLE CM55

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.

INCHES	MM
.003	.08
.010	.25
.02	.51
.03	.76
.25	6.35
.47	11.94
1.250	31.75
1.31	33.27
1.75	44.45



NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Approximate weight 1.02 ounces.

FSC 5910

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REQUIREMENTS:

Dimensions: See figure.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
N	-55° C to +85° C
O	-55° C to +125° C
P	-55° C to +150° C

Vibration grade: 3 (10 to 2,000 cps).

Characteristic:

Symbol	Temperature coefficient P/M/° C	Capacitance drift
B	Not specified	Not specified
C	-200 to +200	±(0.5 percent +0.1 pf)
D	-100 to ±100	±(0.3 percent +0.1 pf)
E	-20 to +100	±(0.1 percent +0.1 pf)

Dielectric withstanding voltage:
Method 301 of MIL-STD-202.
200 percent of dc working voltage for not less than 1 second nor more than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202:
Condition B (50,000 ft).
Test potential - 150 percent of rated dc voltage for not less than 1 second nor more than 5 seconds.

Insulation resistance:
Method 302 of MIL-STD-202, condition A.
At 25° C - 10,000 megohms minimum.
At 85° C - 4,000 megohms minimum.
At 125° C - 1,500 megohms minimum.
At 150° C - 750 megohms minimum.

Dissipation factor, maximum:
See MIL-C-5.

Capacitance:
Method 305 of MIL-STD-202.

Vibration:
Method 204 of MIL-STD-202, condition D (20 G).

Temperature cycling:
Method 102 of MIL-STD-202, condition D.

Immersion:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±4 percent of nominal value or 1 pf, whichever is greater, for characteristics B and C; ±2 percent of nominal value or 1 pf, whichever is greater, for characteristics D and E.

Shock (method impact):
Method 205 of MIL-STD-202, condition C (50 G).

Moisture resistance:
Method 106 of MIL-STD-202.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±4 percent of nominal value or 1 pf, whichever is greater for characteristics B and C; ±2 percent of nominal value or 1 pf, whichever is greater, for characteristics D and E.

Temperature coefficient and capacitance drift:
Not specified.

Low temperature storage and life:
48 hours at $-55\ ^{+0°}_{-2}$ C, immediately followed by qualification or performance test, as applicable.
Qualification test - 2,000 hours at the applicable high test temperature with 150 percent of dc working voltage applied.
Dielectric withstanding voltage - Initial requirements.
Insulation resistance - Initial requirements.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±5 percent of nominal value of 1 pf, whichever is greater, for characteristic B and C; ±2 percent of nominal value or 1 pf, whichever is greater, for characteristics D and E.
Performance chect - Conditions and requirements in accordance with qualification test except test duration shall be 250 hours.
Continuation test - Units that have been subjected to the 250 hour test shall be continued for an additional 1,750 hours and meet the requirements for the qualification test.

Qualification inspection:
Qualification inspection (Group C) is required.

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TABLE I. Style CM55.

Type 1/ designation	Capacitance (pf)	DC working voltage (volts)	Operating temperature range	Characteristics available capacitance tolerances	
				±2%(G)	±5%(J)
CM55BK220J-3	22	2,500	N, O, P		B
CM55BK240J-3	24	2,500	N, O, P		B
CM55BK270J-3	27	2,500	N, O, P		B
CM55BK300J-3	30	2,500	N, O, P		B
CM55BK330J-3	33	2,500	N, O, P		B
CM55BK360J-3	36	2,500	N, O, P		B
CM55BK390J-3	39	2,500	N, O, P		B
CM55BK430J-3	43	2,500	N, O, P		B
CM55BK470J-3	47	2,500	N, O, P		B
CM55BK510--3	51	2,500	N, O, P	B	B
CM55BK560--3	56	2,500	N, O, P	B	B
CM55BK620--3	62	2,500	N, O, P	B	B
CM55BK680--3	68	2,500	N, O, P	B	B
CM55BK750--3	75	2,500	N, O, P	B	B
CM55BK820--3	82	2,500	N, O, P	B	B
CM55BK910--3	91	2,500	N, O, P	B	B
CM55-K101--3	100	2,500	N, O, P	C, D	C, D
CM55-K111--3	110	2,500	N, O, P	C, D	C, D
CM55-K121--3	120	2,500	N, O, P	C, D	C, D
CM55-K131--3	130	2,500	N, O, P	C, D	C, D
CM55-K151--3	150	2,500	N, O, P	C, D	C, D
CM55-K161--3	160	2,500	N, O, P	C, D	C, D
CM55-K181--3	180	2,500	N, O, P	C, D	C, D
CM55-K201--3	200	2,500	N, O, P	C, D	C, D
CM55-K221--3	220	2,500	N, O, P	C, D	C, D
CM55-K241--3	240	2,500	N, O, P	C, D	C, D
CM55-K271--3	270	2,500	N, O, P	C, D	C, D
CM55-K301--3	300	2,500	N, O, P	C, D	C, D
CM55-K331--3	330	2,500	N, O, P	C, E	C, E
CM55-K361--3	360	2,500	N, O, P	C, E	C, E
CM55-K391--3	390	2,500	N, O, P	C, E	C, E
CM55-K431--3	430	2,500	N, O, P	C, E	C, E
CM55-K471--3	470	2,500	N, O, P	C, E	C, E
CM55-K511--3	510	2,500	N, O, P	C, E	C, E
CM55-K561--3	560	2,500	N, O, P	C, E	C, E
CM55-K621--3	620	2,500	N, O, P	C, E	C, E
CM55-K681--3	680	2,500	N, O, P	C, E	C, E
CM55-K751--3	750	2,500	N, O, P	C, E	C, E
CM55-K821--3	820	2,500	N, O, P	C, E	C, E
CM55-K911--3	910	2,500	N, O, P	C, E	C, E
CM55-K102--3	1000	2,500	N, O, P	C, E	C, E
CM55-K112--3	1100	2,500	N, O, P	C, E	C, E
CM55-K122--3	1200	2,500	N, O, P	C, E	C, E
CM55-K132--3	1300	2,500	N, O, P	C, E	C, E
CM55-K152--3	1500	2,500	N, O, P	C, E	C, E
CM55-K162--3	1600	2,500	N, O, P	C, E	C, E
CM55-K182--3	1800	2,500	N, O, P	C, E	C, E
CM55-K202--3	2000	2,500	N, O, P	C, E	C, E
CM55-K222--3	2200	2,500	N, O, P	C, E	C, E
CM55-K242--3	2400	2,500	N, O, P	C, E	C, E
CM55-K272--3	2700	2,500	N, O, P	C, E	C, E
CM55-K302--3	3000	2,500	N, O, P	C, E	C, E
CM55-K332--3	3300	2,500	N, O, P	C, E	C, E
CM55-K362--3	3600	2,500	N, O, P	C, E	C, E

See footnote at end of table.

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TABLE I. Style CM55 - Continued

Type 1/ designation	Capacitance (pf)	DC working voltage (volts)	Operating temperature range	Characteristics available capacitance tolerances	
				±2%(G)	±5%(J)
CM55-K392--3	3900	2,500	N, O, P	C, E	C, E
CM55-K432--3	4300	2,500	N, O, P	C, E	C, E
CM55-K472--3	4700	2,500	N, O, P	C, E	C, E
CM55-G512--3	5100	1,200	N, O, P	C, E	C, E
CM55-G562--3	5600	1,200	N, O, P	C, E	C, E
CM55-G622--3	6200	1,200	N, O, P	C, E	C, E
CM55-G682--3	6800	1,200	N, O, P	C, E	C, E
CM55-G752--3	7500	1,200	N, O, P	C, E	C, E
CM55-G822--3	8200	1,200	N, O, P	C, E	C, E
CM55-G912--3	9100	1,200	N, O, P	C, E	C, E
CM55-G103--3	10,000	1,200	N, O, P	C, E	C, E
CM55-G113--3	11,000	1,200	N, O, P	C, E	C, E
CM55-G123--3	12,000	1,200	N, O, P	C, E	C, E
CM55-GJ33--3	13,000	1,200	N, O, P	C, E	C, E
CM55-E153--3	15,000	600	N, O, P	C, E	C, E
CM55-E163--3	16,000	600	N, O, P	C, E	C, E
CM55-E183--3	18,000	600	N, O, P	C, E	C, E
CM55-E203--3	20,000	600	N, O, P	C, E	C, E
CM55-E223--3	22,000	600	N, O, P	C, E	C, E
CM55-E243--3	24,000	600	N, O, P	C, E	C, E
CM55-E273--3	27,000	600	N, O, P	C, E	C, E
CM55-E303--3	30,000	600	N, O, P	C, E	C, E
CM55-E333--3	33,000	600	N, O, P	C, E	C, E

1/ Complete type designation will include additional symbols to indicate characteristic, capacitance tolerance, and temperature range.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85

User activities:
Army -
Navy -
Air Force - 14, 19

Preparing activity:
Army - EL

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT [_] SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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